

crash-如何找结构体的数据

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练习:

全局变量解析:

直接 p <全局变量的变量名>

局部变量解析:

1、从案例1找出KdumpLkval_st stKdLkval

```
typedef struct kdump_localval {
    char cName[NAMELEN];
    int iNumber;
} KdumpLkval_st;
```

1: #5 [ffff800011b1bd30] kdump_proc_write at ffff800008c40268 [hello]

sp: ffff800011b1bd30===》栈顶地址

2、查看字符串类型局部变量:

C代码:

```
int kdump_stack_test(int num, KdumpStack_st *pstKdStack, KdumpLkval_st *pstKdLkval)
{
    int iCycle = 0;
    char cNodeName[NAMELEN] = {0};
    KdumpNode_st *pstKdNode = NULL;
    /* 链表查看 */
    for(iCycle = 0; iCycle < num; iCycle++) {
        pstKdNode = kzalloc(sizeof(KdumpNode_st), GFP_KERNEL);
        snprintf(cNodeName, NAMELEN, "node_%d", iCycle);
        strncpy(pstKdNode->cName, cNodeName, NAMELEN);
        pstKdNode->iNumber = iCycle;

        list_add_tail(&pstKdNode->list, &pstKdStack->list);
    }

    /* 栈中结构体变量 */
    pstKdLkval->iNumber = 101;
    /* 堆中结构体变量 */
    gpstKdLkval->iNumber = 201;
    /* 全局结构体变量 */
    pstKdStack->printk();

    return 0;
}
```

查找:

char cNodeName[NAMELEN] = {0};

step1: 反汇编, 进入kdump_stack_test函数

根据x0和x1传参的地址来找cNodeName:

```
    snprintf(cNodeName, NAMELEN, "node_%d", iCycle);
    strncpy(pstKdNode->cName, cNodeName, NAMELEN);
0xffff800008c400fc <kdump_stack_test+108>:    mov     x19, x0
0xffff800008c40100 <kdump_stack_test+112>:    mov     w3, w20
0xffff800008c40104 <kdump_stack_test+116>:    mov     x2, x22
0xffff800008c40108 <kdump_stack_test+120>:    mov     x1, #0x14                                // #20
0xffff800008c4010c <kdump_stack_test+124>:    add     x0, sp, #0x54 ===》找到kdump_stack_test堆栈+0x54就是字符串指针地址
0xffff800008c40110 <kdump_stack_test+128>:    bl      0xffff800010c1f7c4 <snprintf>
```

step2: 计算地址:

```
#3 [ffff800011b1bcb0] kdump_printk at ffff800008c403d8 [hello]
#4 [ffff800011b1bcc0] kdump_stack_test at ffff800008c4016c [hello] ===》找到堆栈地址ffff800011b1bcc0
#5 [ffff800011b1bd30] kdump_proc_write at ffff800008c40268 [hello]
#6 [ffff800011b1bd80] proc_wrbuff_write at ffff800008c40378 [hello]
字符串指针地址就是ffff800011b1bcc0+0x54=0xFFFF800011B1BD14
```

step3: 读出字符串:

rd命令	<地址>	<以多大的长度显示>	<显示个数>
crash803> rd	0xFFFF800011B1BD14	-8	10
ffff800011b1bd14: 6e 6f 64 65 5f 33 00 00 00 00			node_3....

3、查看全局变量:

p <变量名>

例:

```
crash803> p stKdumpStack
stKdumpStack = $1 = {
  cName = "StTest\000\000\000\000\000\000\000\000\000\000\000\000",
  iNumbers = 4,
  list = {
    next = 0xffff00001b6b6698,
    prev = 0xffff00001b6b6b98
  },
  printk = 0xffff800008c403a8 <kdump_printk>
}
crash803> p gpstKdLkval
gpstKdLkval = $2 = (KdumpLkval_st *) 0xffff00001b6b6200
crash803> p *gpstKdLkval
$3 = {
  cName = "heap val\000\000\000\000\000\000\000\000\000\000\000",
  iNumber = 201
}
```

#4 [ffff800011b1bcc0] kdump_stack_test at ffff800008c4016c [hello]